

Assignment: Zomato Restaurant Data Analysis using Python

In this assignment, you will perform **Exploratory Data Analysis (EDA)** on the provided **Zomato Restaurant Dataset** using Python libraries such as **Pandas, NumPy, Matplotlib, and Seaborn**.

Your goal is to understand the dataset, perform basic data cleaning, generate statistical insights, and create meaningful visualizations.

This assignment will help you understand:

- Loading and exploring real-world datasets
- Data cleaning and preprocessing
- Basic statistical analysis
- Data visualization
- Drawing business insights from data

Dataset Description

The dataset contains information about restaurants listed on Zomato, including restaurant details, location, cuisines, ratings, pricing, and customer engagement.

Some important columns include:

Column	Description
Restaurant Name	Name of the restaurant
City	City where the restaurant is located
Cuisines	Type of cuisines served
Average Cost for two	Estimated cost for two people
Has Online delivery	Whether online delivery is available
Has Table booking	Whether table booking is available
Price range	Restaurant price category
Aggregate rating	Overall customer rating
Rating text	Rating category
Votes	Number of customer votes

Assignment Tasks

You must complete the following stages.

Part 1 — Data Understanding & Exploration

Perform basic data exploration and preprocessing.

Tasks

1. Load the dataset using Pandas.
2. Explore the dataset structure:
 - a. Shape of the dataset
 - b. Column names
 - c. Data types
 - d. Missing values
3. Identify duplicate records (if any).
4. Convert the "**Average Cost for two**" column into an appropriate numeric data type (if required).
5. Perform basic statistical analysis using appropriate Pandas functions.
6. Answer the following questions:
 - a. How many unique cities are present in the dataset?
 - b. Which city has the highest number of restaurants?
 - c. Which cuisine appears most frequently?

Deliverables

- Summary of dataset
- Missing value analysis
- Data type changes performed
- Key observations from the dataset

Part 2 — Data Analysis

Analyze the dataset to answer the following questions.

Tasks

1. Find the average restaurant rating.
2. Identify the restaurant with the highest number of votes.
3. Find the average cost for two across all restaurants.
4. Compare the average ratings of restaurants that:
 - a. Have Online Delivery
 - b. Do Not Have Online Delivery
5. Compare the average ratings of restaurants that:
 - a. Have Table Booking

- b. Do Not Have Table Booking
6. Identify the top 10 cities with the highest number of restaurants.

Deliverables

- Summary statistics
- Business insights based on the analysis

Part 3 — Data Visualization

Create appropriate visualizations using **Matplotlib** and/or **Seaborn**.

Tasks

Create the following charts:

1. **Bar Chart**
 - a. Top 10 cities with the highest number of restaurants.
2. **Histogram**
 - a. Distribution of Aggregate Ratings.
3. **Bar Chart**
 - a. Number of restaurants offering Online Delivery (Yes/No).
4. **Bar Chart**
 - a. Top 10 most common cuisines.
5. **Line Chart**
 - a. Average restaurant rating by Price Range.

Deliverables

- Well-labeled charts with titles and axis labels
- A short interpretation (2–3 lines) for each visualization

Part 4 — Insights & Conclusion

Based on your analysis, answer the following:

1. Which city has the highest restaurant presence?
2. What is the most popular cuisine?
3. Do restaurants with Online Delivery generally have higher ratings?
4. Do restaurants offering Table Booking receive better ratings?
5. What did you learn from analyzing this dataset?

Expected Learning Outcomes

At the end of this assignment, learners should be able to:

Outcome 1

Load and explore a real-world dataset using Pandas.

Outcome 2

Identify missing values, duplicate records, and perform basic data cleaning.

Outcome 3

Perform descriptive statistical analysis using Python.

Outcome 4

Create basic visualizations using Matplotlib and Seaborn.

Outcome 5

Interpret data and communicate meaningful business insights.

Documentation Requirements

Students must submit a short report (2–3 pages) containing:

1. Dataset Overview

- Description of the dataset
- Number of rows and columns

2. Data Cleaning

- Missing values identified
- Data type changes performed
- Duplicate records (if any)

3. Data Analysis

- Statistical summary

- Answers to the analysis questions
- Key observations

4. Data Visualizations

- All required charts
- Brief explanation of each visualization

5. Conclusion

- Final insights
- Business observations from the dataset

Submission Requirements

Students must submit:

1. Python Notebook (.ipynb)
2. Documentation Report (PDF)
3. Output screenshots and visualizations